

Lesson Notes – Drawing Shapes

Lesson Introduction

In Scratch, you can use the **Pen blocks** to make sprites **draw shapes** and patterns on the stage.

- Just like drawing on paper, you can also draw on the computer screen in Scratch using the pen feature. You can create colorful patterns, shapes, and drawings.
- Drawing shapes helps learners **combine motion, loops, and creativity**.
- You can draw **squares, triangles, circles, and other patterns** using code instead of drawing by hand.

Learning Objectives

By the end of this lesson, learners will be able to:

1. Use **Pen blocks** to draw on the Scratch stage.
2. Make a sprite **draw basic shapes** like squares, triangles, and circles.
3. Combine **loops and motion blocks** to automate drawing.
4. Change **pen size, color, and effects** for creativity.
5. Create simple **designs and patterns** using code.

Key Vocabulary

- **Pen Down:** Lowers the pen so the sprite draws while moving.
- **Pen Up:** Lifts the pen so the sprite moves without drawing.
- **Set Pen Color:** Changes the color of the line.
- **Set Pen Size:** Changes how thick the line is.
- **Move / Turn:** Used with loops to create shapes.

1. The Pen Block:

The "Pen" blocks in Scratch controls the pen's behavior, such as lifting or lowering it and changing its color.

2. Basic Pen Commands:

pen down: Put the pen on the paper (or screen) so that it can draw.

pen up: Lifts the pen, so it stops drawing.

3. Changing Pen Color:

To change the pen color we use the set pen color to block. You can experiment with different colors.

4. Drawing Shapes:

We can draw basic shapes, like circles and squares, using loops and pen commands.

Step-by-Step Guided Activity

Step 1: Set Up

1. Open Scratch 3 and click **Create**.
2. Add a sprite (e.g., Cat).
3. Add a backdrop if desired (optional).

Step 2: Draw a Square

1. Select the **Pen blocks** category.
2. Add the following code:

```
when green flag clicked
pen down
repeat 4
  move 100 steps
  turn 90 degrees
pen up
```

The sprite will draw a square on the stage.

Step 3: Draw a Triangle

```
when green flag clicked
pen down
repeat 3
  move 100 steps
  turn 120 degrees
pen up
```

The sprite draws a triangle using a loop.

Step 4: Draw a Circle (Approximation)

```
when green flag clicked
pen down
repeat 36
  move 10 steps
  turn 10 degrees
pen up
```

The sprite draws a circle by turning slightly many times.

Step 5: Optional Enhancements

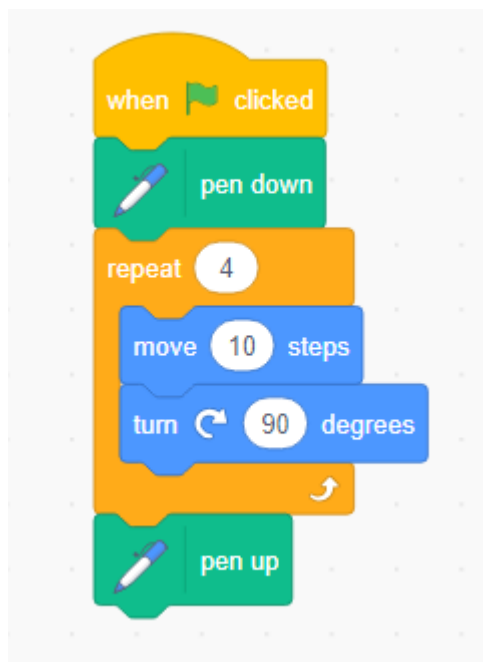
- Change **pen color** for each shape: set pen color to [red].
- Change **pen size**: set pen size to 5.
- Use **repeat loops inside repeat loops** to make patterns or complex designs.
- Add **sounds or costumes** while drawing to make it interactive.

Example 2

How to draw a simple square.

We use the concept of angles in the square which is:

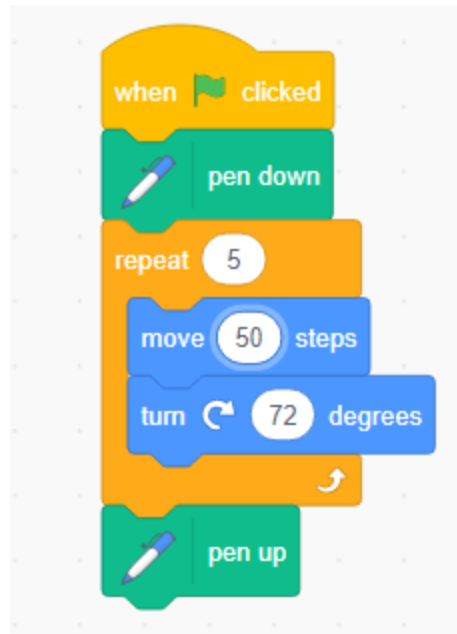
Formula = $360/\text{number of sides of the shape}$.



Example 3

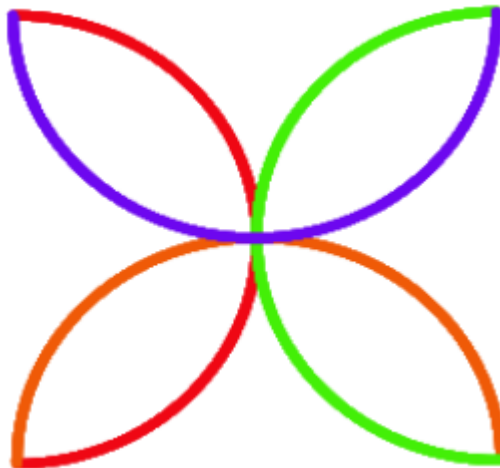
How to draw a pentagon.

This is to emphasize the use of the formula to get the angle in each corner of the shape.



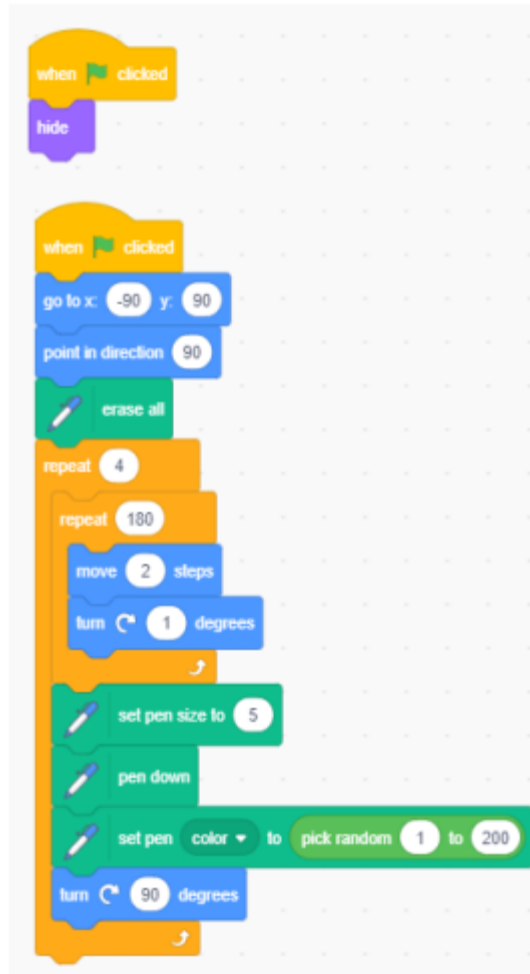
Example 4

Create A four petal Flower



The effect is achieved by making four half circles (180 degrees) and turning right (90 degrees) at the completion of each one.

This is achieved by the following code snippet



Project

Draw a Pentagon, triangle and an octagon.

Write a programme that draws three different shapes that appear at different times at different locations on the Stage.

Hints

Recap the key concepts: pen down, pen up, changing pen color, erase all, set pen size, and drawing shapes.

Children should share their drawings with each other, fostering a sense of accomplishment.

Discussion / Questions

1. What block makes the sprite draw on the stage?
2. How does a loop help in drawing shapes?
3. How do you draw different shapes using the Pen blocks?

4. How can pen color and size make your drawings more creative?
5. Can you combine loops and motion to make patterns?

Learning Outcome

By the end of this lesson, learners will be able to:

- Use **Pen blocks** to draw basic shapes in Scratch.
- Combine **loops and motion** to automate drawing.
- Create **colorful designs and patterns** on the stage.
- Apply coding concepts to **develop creativity and problem-solving skills**.